Cepheus 2.0

RouteGuard

Preemptively detecting supply chain bottlenecks before they cascade.

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Problem Statement & Background

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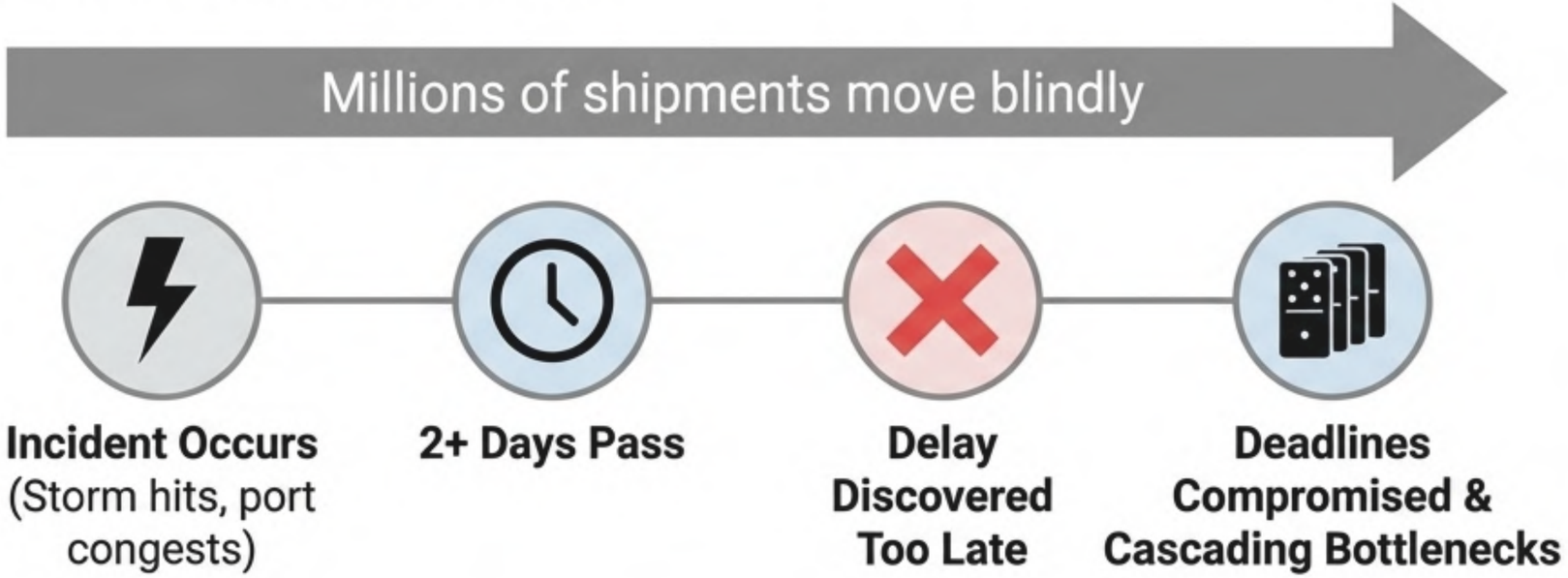


- Supply chain management is currently reactive.
- Incidents like weather or congestion are discovered after they occur.
- This delay leads to missed deadlines and cascading failures.

The industry tracks the past. It cannot predict the future.

Background:

$1.5 Trillion

Yearly global loss from preventable supply chain disruptions.

Traditional tracking networks are highly complex but fundamentally flawed—managers only find out about a port strike or severe weather event after timelines are already ruined.

Solution Overview

- **Continuous Live Monitoring:** Ingests multifaceted transit data (APIs, simulated streams) continuously for real-time visibility.
- **Predictive Disruption Engine:** Utilizes Machine Learning to generate dynamic 0-100 risk scores, identifying potential bottlenecks before they cascade.
- **Smart Rerouting Recommendation:** Provides instant, optimized route adjustments based on time vs. cost, enabling proactive decision-making.

1. Continuous Live Monitoring

Ingesting multifaceted transit data continuously.

2. Predictive Disruption Engine

Generating dynamic 0-100 risk scores before incidents cascade.

3. Smart Rerouting Recommendation

Instantly executing optimized route adjustments based on time vs. cost.

System Architecture: Polyglot Data & ML Processing Stack

Layer 1: External Data Sources

Live Data APIs

- OpenWeatherMap
- TomTom Traffic
- OpenRouteService

Simulated Streams

- Port Congestion
- Historical Delays
- AIS Vessel Data

Layer 2: Core Processing & ML

Python Backend + Feature Engineering

Converts raw data into standardized 0-100 scoring metrics.

Layer 3: Polyglot Persistence

PostgreSQL (Structured)

Stores static users, vessels, ports, shipments.

MongoDB (Time-Series)

Stores high-velocity real-time telemetry & ML logs.

Layer 4: Delivery Interface

FastAPI Backend

React + MapLibre Dashboards pushing live WebSockets updates.

Ecosystem Access: The 4-Role Permission Architecture

Shipper / Sender

Factory or brand sending goods.

- Create new shipments
- View milestone tracking
- Receive automated delay alerts

Logistics Manager (Hero User)

Supply chain manager overseeing all operations.

- Complete live map visibility
- ML feature breakdown & risk score details
- 1-click alternative reroute approval
- Financial impact analytics

Driver / Captain

Carrier physically moving the cargo.

- View assigned routes
- Receive live diversion alerts
- Report localized incidents

Receiver / Consignee

Destination facility.

- Incoming shipment live tracking
- Digital delivery confirmation

Technical Implementation: The Dual-Model ML Pipeline

Weather

Traffic

Port History

Cargo Details

Feature Engineering (Python):
Converts to 0-100 Standardized Scores

Model 1: XGBoost Regressor

Predicts continuous **Risk Score** (0-100). Chosen for speed with mixed tabular data.

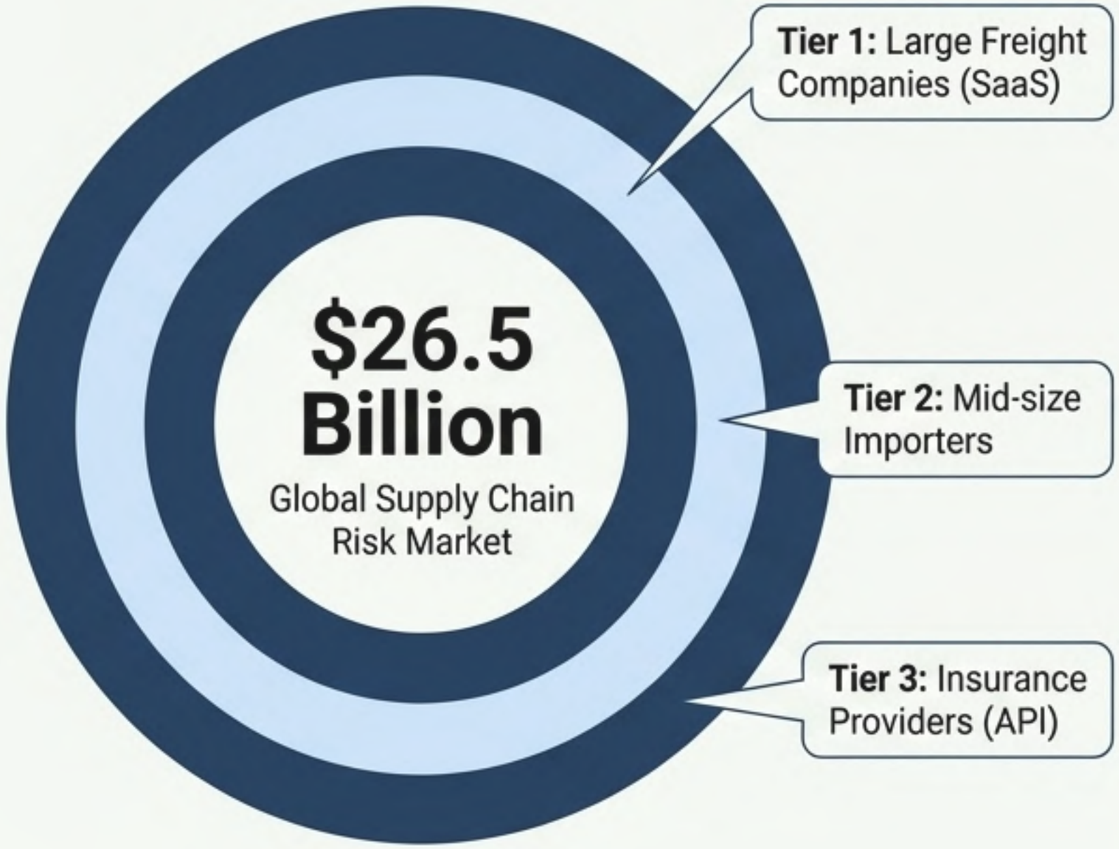
Model 2: Random Forest Regressor

Predicts exact **Delay Duration** in hours based on current route.

Threshold Automation Table

0-30 (Low Risk)	Action: Continue Monitoring
31-60 (Medium Risk)	Action: Alert Logistics Manager
61-80 (High Risk)	Action: Fetch 2 alternative routes via OpenRouteService API
81-100 (Critical Risk)	Action: Demand immediate reroute approval

Market Feasibility & Real-World Impact



The Competitive Advantage

Traditional Tools	RouteGuard
• Track the past • Limited to GPS • Reactive manual changes	• Predict the future • ML-driven multi-source data • 1-click automated rerouting

Impact Scorecard

Financial Impact: Prevents massive late-delivery penalties and reduces insurance premiums.

Environmental Impact: Slashes wasted fuel and emissions through proactive avoidance of idling in congestion.

Future Scope: The Roadmap to a Global Standard

Next 3 Months: Production Ready

Transition to live MarineTraffic API integration.

Launch closed pilot with 5 mid-size logistics companies.

6 Months: Feature Expansion

Integrate live IoT sensor data for sensitive cargo.

Implement Customs Document AI to predict border delays.

12–18 Months: Platform Evolution

Deploy True Autonomous Rerouting based on rulesets.

Introduce Digital Twin supply chain simulation for enterprises.

The Ultimate Vision: To become the underlying intelligence layer for global transit—where no shipment is ever delayed by a predictable event.